Predicting 48-Hour **Extubation Failure** Using Pre-Extubation Clinical Data: A Retrospective MIMIC-IV Study

GROUP 25

Wenhao Duan
Jiajian He
Haoran Zhang
Kun Xing
Zongcheng Jiang

Introduction



Clinical Problem

Early extubation risks failure.
Delay prolongs ventilation.

42.3% vs 14.0%

Hospital mortality with vs without reintubation within 48h.



Current Gap

RSBI alone provides incomplete predictive information.

Multivariable prediction combines clinical measurements



Study Aim

Predict reintubation or in-hospital death within 48h of extubation.

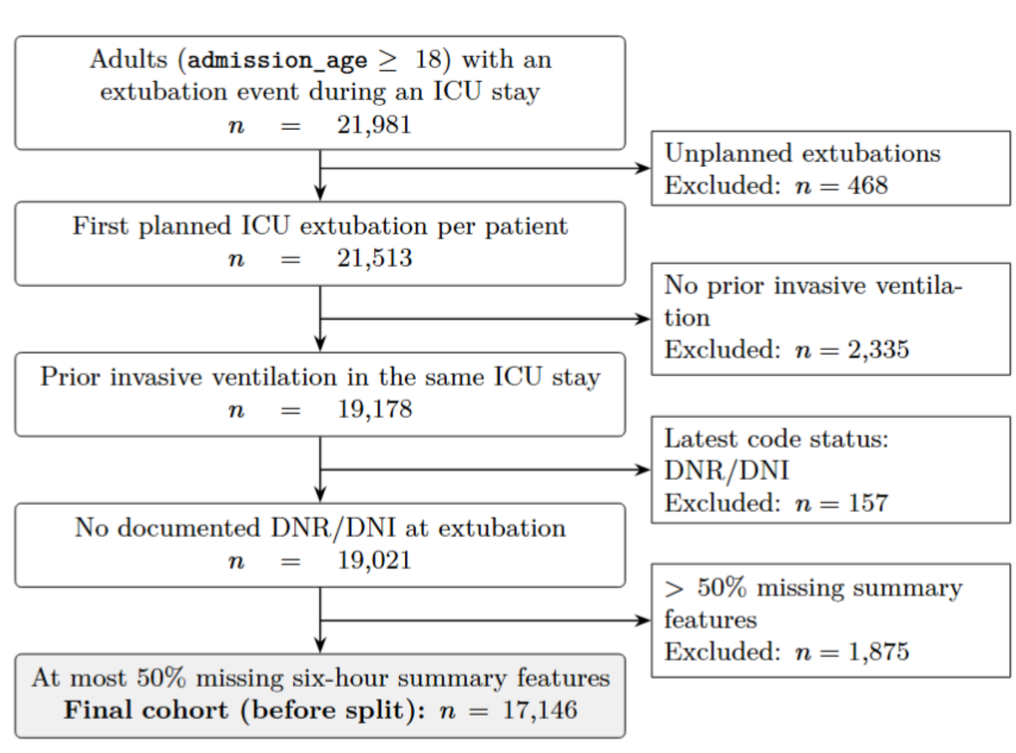
Logistic regression vs XGBoost

RSBI alone provides incomplete predictive information.

Intended use: Inform risk assessment and planning of post-extubation monitoring.

Methodology: Dataset & Data Preprocessing

Cohort Construction



- MIMIC-IV v3.1, extracted using Google BigQuery.
- First planned extubation per adult patient, following invasive ventilation.
- Final cohort: 17,146 patients.
- **Outcome**: reintubation or in-hospital death within 48 hours.

Features

```
heart_rate
mbp

resp_rate

temperature

spo2
respiratory_rate_spontaneous
tidal_volume_spontaneous
tidal_volume_observed
minute_volume
peep

fio2_chartevents
ph
pco2
po2
bicarbonate
lactate
fio2

pao2fio2ratio

RSBI
```

- Prediction window: six hours immediately before extubation.
- 19 physiological summaries: vital signs, ventilator measurements, arterial blood gases, and derived respiratory indices (RSBI and P/F ratio).
- **Static predictors**: age, sex, ICU admission type, and hospital/ICU duration before extubation.
- **Aggregation**: mean of available valid measurements → one row per patient.
- **Missing data**: exclude patients missing >50% of physiological summaries; impute remaining numerical values using training-fold medians.
- **Leakage prevention**: include only measurements recorded and available before extubation.

Methodology: Models



Chance

6 Hour mean RSBI
Clinical baseline

Logistic Regression
Linear interpretation

XGBoost
Nonlinear + interaction

5 outer folds
15493 development patients
compare models

Select threshold:
maximise specificity at ≥80% sensitivity

3 inner folds
tune hyperparameter

→

Outer validation
AUROC

Weighting and calibration

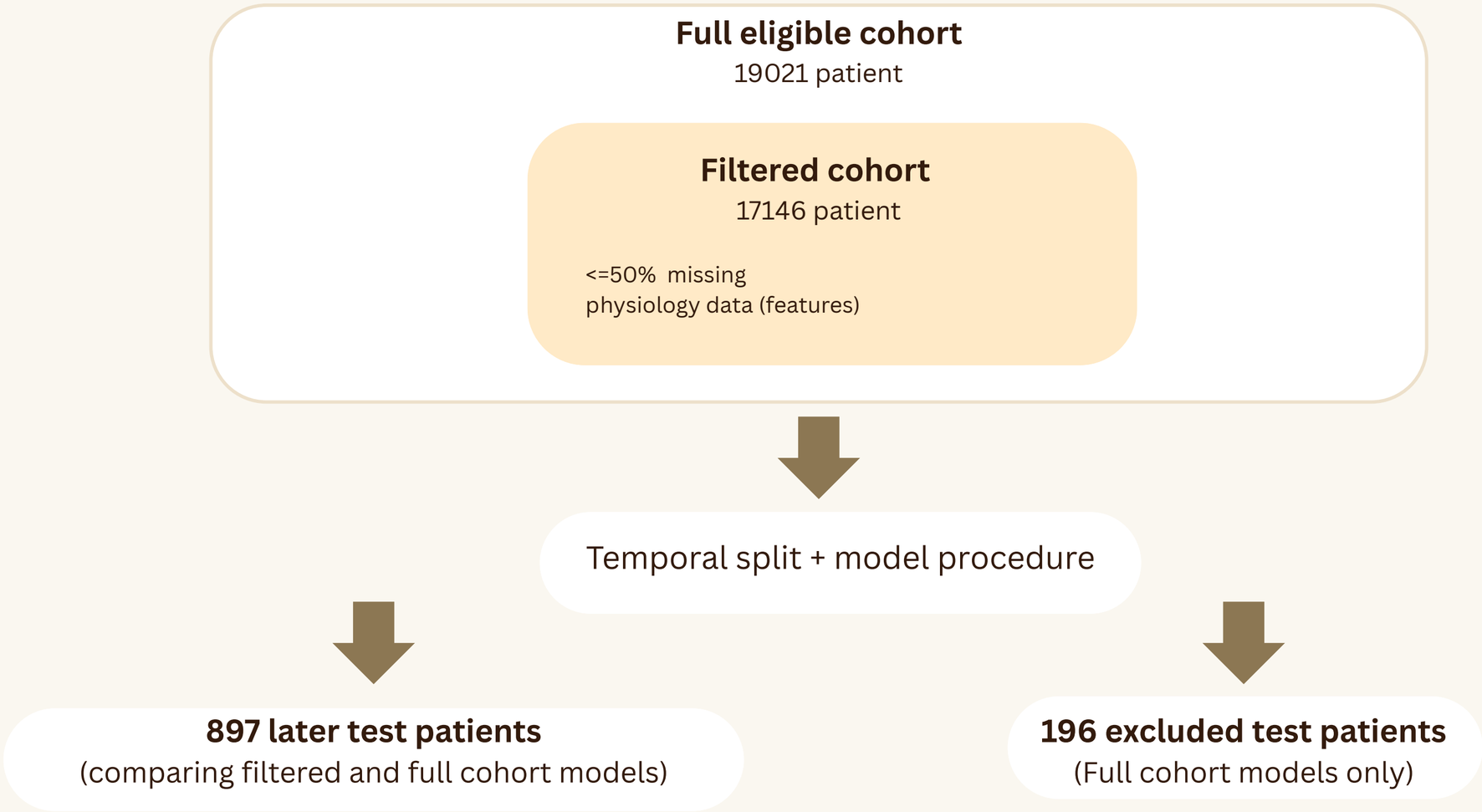
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Held-out test
897 later patients

(starting in 2020 or later)

756 patients excluded cutoff-spanning intervals

Methodology: Sensitivity analysis



Exploratory analysis added after primary test inspection
The primary analysis remained unchanged.

Results: Model Performance

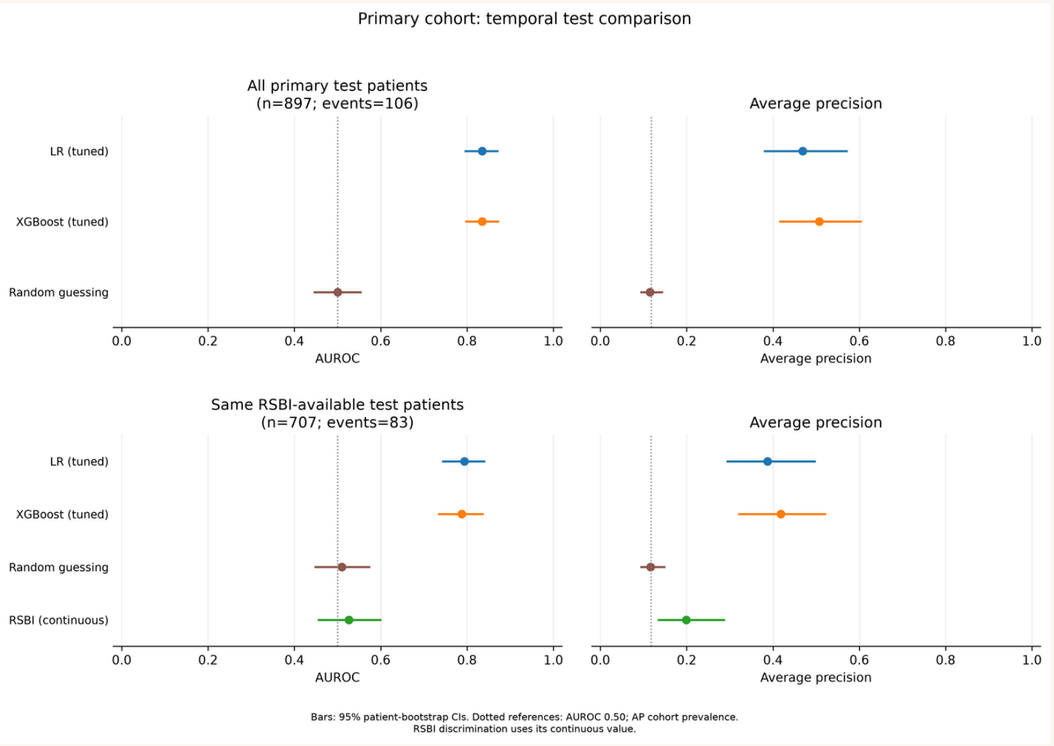
Temporal test cohort: 897 patients, 106 failures (11.8%)

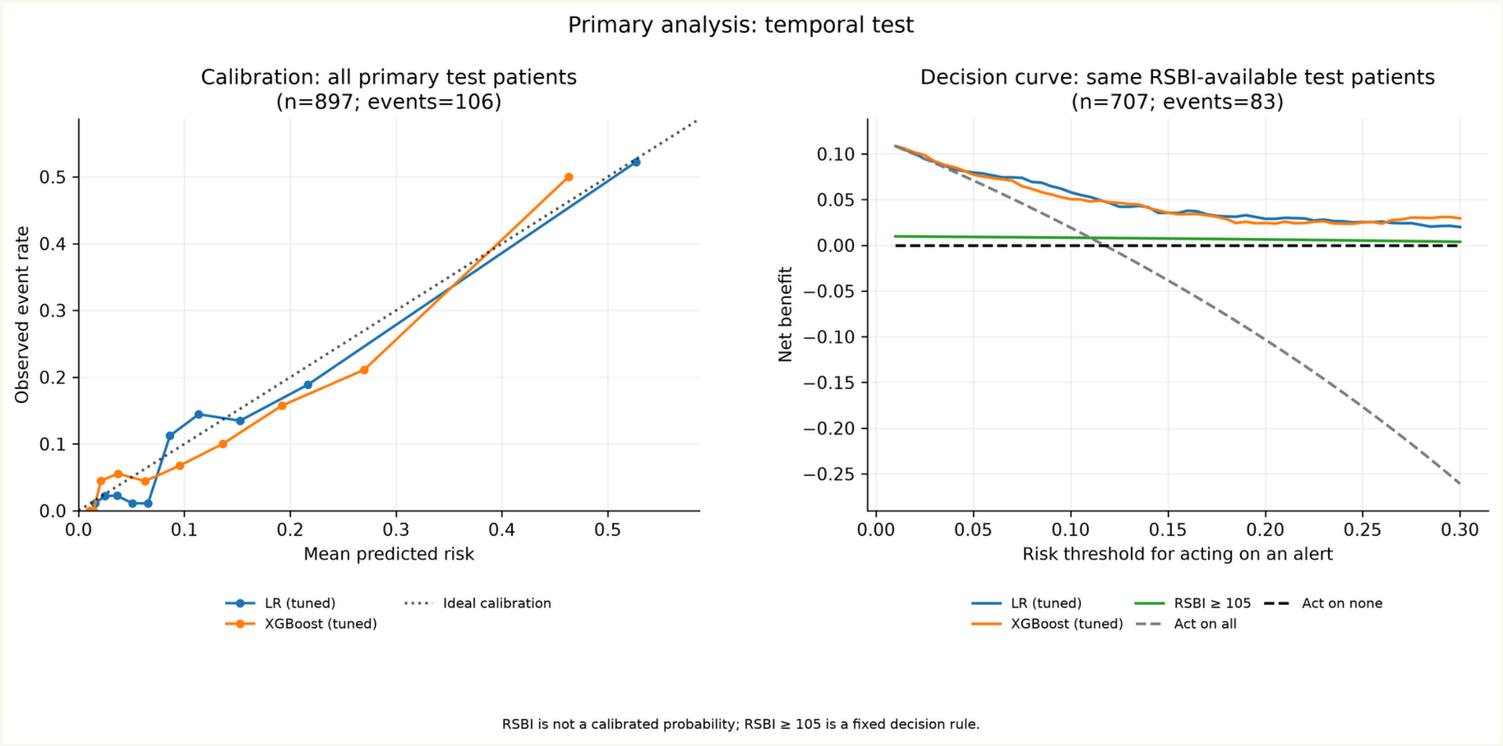
Similar discrimination: AUROC 0.835 for both; AP 0.469 for LR and 0.507 for XGBoost.

Similar probability error: Brier ≈ 0.0805 for both, with slight overall risk overprediction.

Clinical comparison: Higher AUROC estimates than RSBI in 707 matched patients;

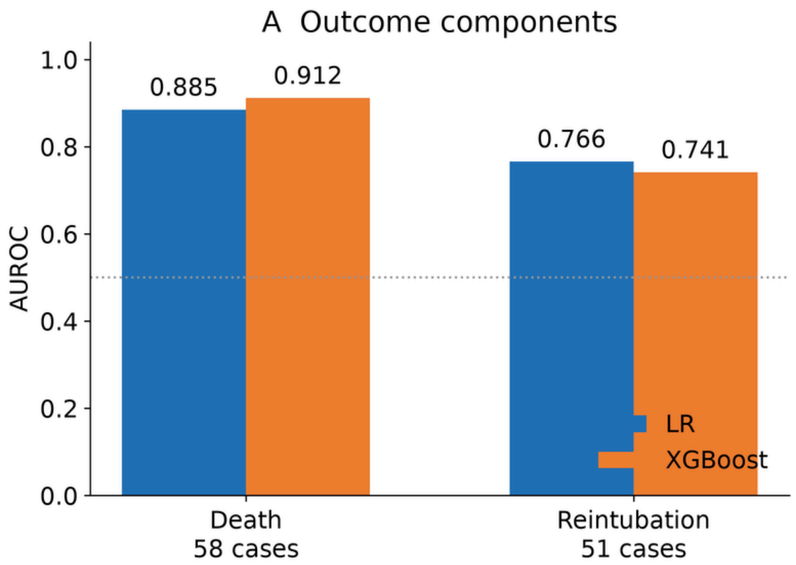
DCA suggests potential net benefit.



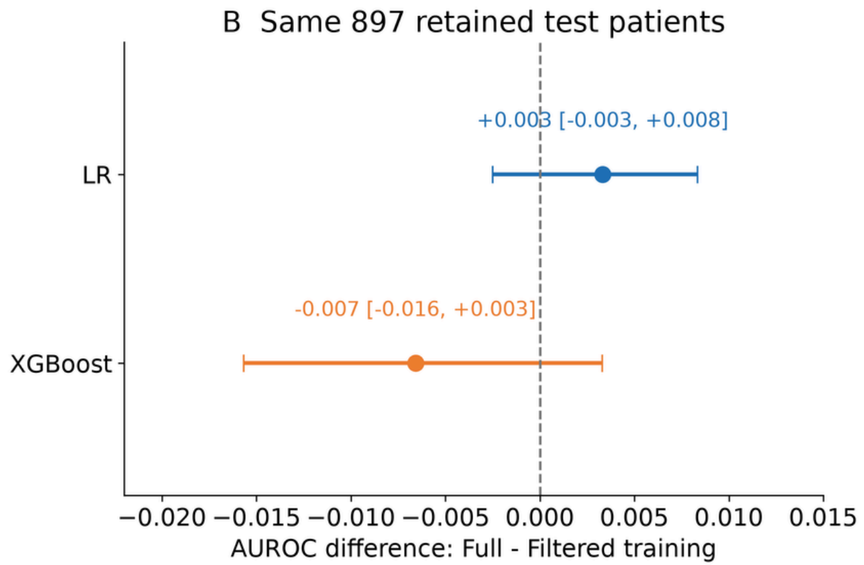


Only tuned LR and XGBoost are shown; untuned models were exploratory comparators.

Results: Error & Sensitivity Analysis



Shared controls: 791 patients with neither outcome.
Different case groups; descriptive comparison only.



Points and bars: paired differences and 95% CIs.
Both AUROC intervals include zero.

Excluded test patients: 57/196 failures (29.1%); Full-trained AUROC: LR 0.710, XGBoost 0.723.

Outcome components: Higher AUROC for death than reintubation in descriptive analyses.

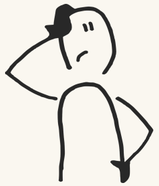
Training cohort: No consistent improvement from Full-cohort training on the same 897 test patients.

Excluded patients: 57/196 failures (29.1%); Full-trained AUROC 0.710 for LR and 0.723 for XGBoost.

Discussion

Interpretation of Findings

- **No ML superiority:** LR and XGBoost achieved identical discrimination **(AUROC 0.835),** outperforming 6h-mean RSBI.
- **Composite caveat:** Performance was highly divergent (Death AUROC ~0.90 vs. Reintubation ~0.74). The model is better at predicting mortality than specific extubation failure.
- **What drives the model?** Top features (ICU hours, ICU type) are non-actionable proxies for illness severity, rather than modifiable pre-extubation physiology.



Clinical Implications

- **Intended Use:** Risk stratification at planned extubation to trigger **enhanced post-extubation monitoring** (e.g., increased RT/nurse reassessment frequency), **NOT** to decide whether to extubate.
- **The Trade-off:** High sensitivity (90.6%) but low PPV (23.2%). In practice, ~4 out of 5 alerts would be false positives.
- **Risk of Deployment:** High false-alert burden may cause **alert fatigue** and increase workload. Decision curve analysis suggests potential net benefit, but prospective feasibility is unproven.

Limitations and Future Work

Key Limitations	Next Steps
Selection bias (Missingness) Excluded 35% of failures **Impact:** Primary results likely optimistic	Test under full cohort & explore multiple imputation to avoid excluding high-risk patients.
Composite Outcome Dissonance AUROC for death (~0.90) >> reintubation (~0.74) **Impact:** Model is driven by mortality, less specific to reintubation	Separate endpoints for future validation. Evaluate dynamic trajectories (trends) rather than just 6h averages.
Confounded Feature Importance Model relies heavily on ICU duration & type **Impact:** These are non-actionable proxies for illness severity	Incorporate physiological time-series and conduct external validation to test generalisability.
Untested Clinical Utility High sensitivity (90.6%) but PPV only 23.2%. **Impact:** High risk of alert fatigue; workflow feasibility is unproven.	Conduct silent prospective trials to assess workload and clinician behavior before real implementation.

Conclusion

1. **Predictive Performance**
   - Pre-extubation data enabled retrospective risk prediction (AUROC 0.835), outperforming 6h-mean RSBI.
   - Complex ML (XGBoost) showed **no advantage** over Logistic Regression.
2. **Clinical Translation**
   - The model serves as an **advisory tool** for post-extubation monitoring, not a gatekeeper for extubation decisions.
   - At the chosen threshold, high false alerts (PPV 23.2%) necessitate careful consideration of ICU workload.
3. **Path Forward**
   - Missingness-related selection bias highlights the need for **external validation**.
   - Future steps require **prospective silent trials** to evaluate real-world clinical utility and workflow integration.



Thank You!

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Jiajian he

AI declaration

We declare the use of ChatGPT[openai.com] in checking the structure of the slide